choose the tag q_i to be the midpoint of the interval I_i. Since both t_i and q_i belong to this interval, we have $|t_i - q_i| < \delta$. Using the Triangle Inequality, we deduce

$$\left|S(h;\dot{\mathcal{P}}) - S(h;\ \dot{\mathcal{Q}})\right| = \left|\sum_{i=1}^{n} t_i(x_i - x_{i-1}) - \sum_{i-1}^{n} q_i(x_i - x_{i-1})\right|$$

$$\leq \sum_{i=1}^{n} |t_i - q_i|(x_i - x_{i-1}) < \delta \sum_{i=1}^{n} (x_i - x_{i-1}) = \delta(x_n - x_0) = \delta.$$

Since $S(h;\ \dot{\mathcal{Q}}) = \frac{1}{2}$, we infer that if $\dot{\mathcal{P}}$ is any tagged partition with $||\dot{\mathcal{P}}|| < \delta$, then

$$\left|S(h;\ \dot{\mathcal{P}}) - \frac{1}{2}\right| < \delta.$$

Therefore we are led to take $\delta_\varepsilon \leq \varepsilon$. If we choose $\delta_\varepsilon := \varepsilon$, we can retrace the argument to conclude that $h \in \mathcal{R}[0,\ 1]$ and $\int_0^1 h = \int_0^1 x\,dx = \frac{1}{2}$.

(d) Let $G(x) := 1/n$ for $x = 1/n$ $(n \in \mathbb{N})$, and $G(x) := 0$ elsewhere on $[0,\ 1]$.

Given $\varepsilon > 0$, the set $E := \{x : G(x) \geq \varepsilon\}$ is a finite set. (For example, if $\varepsilon = 1/10$, then $E = \{1,\ 1/2,\ 1/3, \ldots, 1/10\}$.) If n is the number of points in E, we allow for the possibility that a tag may be counted twice if it is an endpoint and let $\delta := \varepsilon/2n$. For a given tagged partition $\dot{\mathcal{P}}$ such that $||\dot{\mathcal{P}}|| < \delta$, we let $\dot{\mathcal{P}}_0$ be the subset of $\dot{\mathcal{P}}$ with all tags outside of E and let $\dot{\mathcal{P}}_1$ be the subset of $\dot{\mathcal{P}}$ with one or more tags in E. Since $G(x) < \varepsilon$ for each x outside of E and $G(x) \leq 1$ for all x in [0,1], we get

$$0 \leq S(G;\ \dot{\mathcal{P}}) = S(G;\ \dot{\mathcal{P}}_0) + S(G;\ \dot{\mathcal{P}}_1) < \varepsilon + (2n)\delta = 2\varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we conclude that G is Riemann integrable with integral equal to zero. □

Some Properties of the Integral

The difficulties involved in determining the value of the integral and of δ_ε suggest that it would be very useful to have some general theorems. The first result in this direction enables us to form certain algebraic combinations of integrable functions.

7.1.5 Theorem *Suppose that f and g are in $\mathcal{R}[a,\ b]$. Then:*

(a) If $k \in \mathbb{R}$, *the function kf is in $\mathcal{R}[a,\ b]$ and*

$$\int_a^b kf = k\int_a^b f.$$

(b) *The function $f + g$ is in $\mathcal{R}[a,\ b]$ and*

$$\int_a^b (f+g) = \int_a^b f + \int_a^b g.$$

(c) *If $f(x) \leq g(x)$ for all $x \in [a, b]$, then*

$$\int_a^b f \leq \int_a^b g.$$

Proof. If $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is a tagged partition of $[a, b]$, then it is an easy exercise to show that

$$S(kf; \dot{\mathcal{P}}) = kS(f; \dot{\mathcal{P}}), \qquad S(f+g; \dot{\mathcal{P}}) = S(f; \dot{\mathcal{P}}) + S(g; \dot{\mathcal{P}}),$$
$$S(f; \dot{\mathcal{P}}) \leq S(g; \dot{\mathcal{P}}).$$

We leave it to the reader to show that the assertion (a) follows from the first equality. As an example, we will complete the proofs of (b) and (c).

Given $\varepsilon > 0$, we can use the argument in the proof of the Uniqueness Theorem 7.1.2 to construct a number $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is any tagged partition with $||\dot{\mathcal{P}}|| < \delta_\varepsilon$, then both

$$\left|S(f; \dot{\mathcal{P}}) - \int_a^b f\right| < \varepsilon/2 \qquad \text{and} \qquad \left|S(g; \dot{\mathcal{P}}) - \int_a^b g\right| < \varepsilon/2. \tag{5}$$

To prove (b), we note that

$$\begin{aligned}\left|S(f+g; \dot{\mathcal{P}}) - \left(\int_a^b f + \int_a^b g\right)\right| &= \left|S(f; \dot{\mathcal{P}}) + S(g; \dot{\mathcal{P}}) - \int_a^b f - \int_a^b g\right| \\ &\leq \left|S(f; \dot{\mathcal{P}}) - \int_a^b f\right| + \left|S(g; \dot{\mathcal{P}}) - \int_a^b g\right| \\ &< \varepsilon/2 + \varepsilon/2 = \varepsilon.\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, we conclude that $f + g \in \mathcal{R}[a, b]$ and that its integral is the sum of the integrals of f and g.

To prove (c), we note that the Triangle Inequality applied to (5) implies

$$\int_a^b f - \varepsilon/2 < S(f; \dot{\mathcal{P}}) \qquad \text{and} \qquad S(g; \dot{\mathcal{P}}) < \int_a^b g + \varepsilon/2.$$

If we use the fact that $S(f; \dot{\mathcal{P}}) \leq S(g; \dot{\mathcal{P}})$, we have

$$\int_a^b f \leq \int_a^b g + \varepsilon.$$

But, since $\varepsilon > 0$ is arbitrary, we conclude that $\int_a^b f \leq \int_a^b g$. Q.E.D.

Boundedness Theorem

We now show that an unbounded function cannot be Riemann integrable.

7.1.6 Theorem *If $f \in \mathcal{R}[a, b]$, then f is bounded on $[a, b]$.*

Proof. Assume that f is an unbounded function in $\mathcal{R}[a, b]$ with integral L. Then there exists $\delta > 0$ such that if $\dot{\mathcal{P}}$ is any tagged partition of $[a, b]$ with $||\dot{\mathcal{P}}|| < \delta$, then we have $\left|S(f; \dot{\mathcal{P}}) - L\right| < 1$, which implies that

$$\left|S(f; \dot{\mathcal{P}})\right| < |L| + 1. \tag{6}$$

Now let $\mathcal{Q} = \{[x_{i-1}, x_i]\}_{n=1}^n$ be a partition of $[a, b]$ with $||\mathcal{Q}|| < \delta$. Since $|f|$ is not bounded on $[a, b]$, then there exists at least one subinterval in $\mathcal{Q}$, say $[x_{k-1}, x_k]$, on which $|f|$ is not bounded—for, if $|f|$ is bounded on each subinterval $[x_{i-1}, x_i]$ by M_i, then it is bounded on $[a, b]$ by $\max\{M_1, \ldots, M_n\}$.

We will now pick tags for $\mathcal{Q}$ that will provide a contradiction to (6). We tag $\mathcal{Q}$ by $t_i := x_i$ for $i \neq k$ and we pick $t_k \in [x_{k-1}, x_k]$ such that

$$|f(t_k)(x_k - x_{k-1})| > |L| + 1 + \left|\sum_{i \neq k} f(t_i)(x_i - x_{i-1})\right|.$$

From the Triangle Inequality (in the form $|A + B| \geq |A| - |B|$), we have

$$\left|S(f; \dot{\mathcal{Q}})\right| \geq |f(t_k)(x_k - x_{k-1})| - \left|\sum_{i \neq k} f(t_i)(x_i - x_{i-1})\right| > |L| + 1,$$

which contradicts (6). Q.E.D.

We will close this section with an example of a function that is discontinuous at every rational number and is not monotone, but is Riemann integrable nevertheless.

7.1.7 Example We consider Thomae's function $h : [0, 1] \to \mathbb{R}$ defined, as in Example 5.1.6(h), by $h(x) := 0$ if $x \in [0, 1]$ is irrational, $h(0) := 1$ and by $h(x) := 1/n$ if $x \in [0, 1]$ is the rational number $x = m/n$ where $m, n \in \mathbb{N}$ have no common integer factors except 1. It was seen in 5.1.6(h) that h is continuous at every irrational number and discontinuous at every rational number in [0, 1]. See Figure 5.1.2. We will now show that $h \in \mathcal{R}[0, 1]$.

For $\varepsilon > 0$, the set $E := \{x \in [0, 1] : h(x) \geq \varepsilon/2\}$ is a finite set. (For example, if $\varepsilon/2 = 1/5$, then there are eleven values of x such that $h(x) \geq 1/5$, namely, $E = \{0, 1, 1/2, 1/3, 2/3, 1/4, 3/4, 1/5, 2/5, 3/5, 4/5\}$. (Sketch a graph.) We let n be the number of elements in E and take $\delta := \varepsilon/(4n)$. If $\dot{\mathcal{P}}$ is a given tagged partition such that $||\dot{\mathcal{P}}|| < \delta$, then we separate $\dot{\mathcal{P}}$ into two subsets. We let $\dot{\mathcal{P}}_1$ be the collection of tagged intervals in $\dot{\mathcal{P}}$ that have their tags in E, and we let $\dot{\mathcal{P}}_2$ be the subset of tagged intervals in $\dot{\mathcal{P}}$ that have their tags elsewhere in [0, 1]. Allowing for the possibility that a tag of $\dot{\mathcal{P}}_1$ may be an endpoint of adjacent intervals, we see that $\dot{\mathcal{P}}_1$ has at most $2n$ intervals and the total length of these intervals can be at most $2n\delta = \varepsilon/2$. Also, we have $0 < h(t_i) \leq 1$ for each tag t_i in $\dot{\mathcal{P}}_1$. Consequently, we have $S(h; \dot{\mathcal{P}}_1) \leq 1 \cdot 2n\delta \leq \varepsilon/2$. For tags t_i in $\dot{\mathcal{P}}_2$, we have $h(t_i) < \varepsilon/2$ and the total length of the subintervals in $\dot{\mathcal{P}}_2$ is clearly less than 1, so that $S(h; \dot{\mathcal{P}}_2) < (\varepsilon/2) \cdot 1 = \varepsilon/2$. Therefore, combining these results, we get

$$0 \leq S(h; \dot{\mathcal{P}}) = S(h; \dot{\mathcal{P}}_1) + S(h; \dot{\mathcal{P}}_2) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we infer that $h \in \mathcal{R}[0, 1]$ with integral 0. □

Exercises for Section 7.1

1. If $I := [0, 4]$, calculate the norms of the following partitions:
   (a) $\mathcal{P}_1 := (0, 1, 2, 4)$, (b) $\mathcal{P}_2 := (0, 2, 3, 4)$,
   (c) $\mathcal{P}_3 := (0, 1, 1.5, 2, 3.4, 4)$, (d) $\mathcal{P}_4 := (0, .5, 2.5, 3.5, 4)$.

2. If $f(x) := x^2$ for $x \in [0, 4]$, calculate the following Riemann sums, where $\dot{\mathcal{P}}_i$ has the same partition points as in Exercise 1, and the tags are selected as indicated.
   (a) $\dot{\mathcal{P}}_1$ with the tags at the left endpoints of the subintervals.
   (b) $\dot{\mathcal{P}}_1$ with the tags at the right endpoints of the subintervals.
   (c) $\dot{\mathcal{P}}_2$ with the tags at the left endpoints of the subintervals.
   (d) $\dot{\mathcal{P}}_2$ with the tags at the right endpoints of the subintervals.

3. Show that $f : [a, b] \to \mathbb{R}$ is Riemann integrable on $[a, b]$ if and only if there exists $L \in \mathbb{R}$ such that for every $\varepsilon > 0$ there exists $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is any tagged partition with norm $||\dot{\mathcal{P}}|| \leq \delta_\varepsilon$, then $\left|S(f; \dot{\mathcal{P}}) - L\right| \leq \varepsilon$.

4. Let $\dot{\mathcal{P}}$ be a tagged partition of $[0, 3]$.
   (a) Show that the union U_1 of all subintervals in $\dot{\mathcal{P}}$ with tags in $[0, 1]$ satisfies $\left[0, 1 - \|\dot{\mathcal{P}}\|\right] \subseteq U_1 \subseteq \left[0, 1 + \|\dot{\mathcal{P}}\|\right]$.
   (b) Show that the union U_2 of all subintervals in $\dot{\mathcal{P}}$ with tags in $[1,2]$ satisfies $\left[1 + \|\dot{\mathcal{P}}\|, 2 - \|\dot{\mathcal{P}}\|\right] \subseteq U_2 \subseteq \left[1 - \|\dot{\mathcal{P}}\|, 2 + \|\dot{\mathcal{P}}\|\right]$.
5. Let $\dot{\mathcal{P}} := \{(I_i, t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$ and let $c_1 < c_2$.
   (a) If u belongs to a subinterval I_i whose tag satisfies $c_1 \leq t_i \leq c_2$, show that $c_1 - \|\dot{\mathcal{P}}\| \leq u \leq c_2 + \|\dot{\mathcal{P}}\|$.
   (b) If $v \in [a, b]$ and satisfies $c_1 + \|\dot{\mathcal{P}}\| \leq v \leq c_2 - \|\dot{\mathcal{P}}\|$, then the tag t_i of any subinterval I_i that contains v satisfies $t_i \in [c_1, c_2]$.
6. (a) Let $f(x) := 2$ if $0 \leq x < 1$ and $f(x) := 1$ if $1 \leq x \leq 2$. Show that $f \in \mathcal{R}[0, 2]$ and evaluate its integral.
   (b) Let $h(x) := 2$ if $0 \leq x < 1$, $h(1) := 3$ and $h(x) := 1$ if $1 < x \leq 2$. Show that $h \in \mathcal{R}[0, 2]$ and evaluate its integral.
7. Use Mathematical Induction and Theorem 7.1.5 to show that if $f_1, \ldots, f_n$ are in $\mathcal{R}[a, b]$ and if $k_1, \ldots, k_n \in \mathbb{R}$, then the linear combination $f = \sum_{i=1}^n k_i f_i$ belongs to $\mathcal{R}[a, b]$ and $\int_a^b f = \sum_{i=1}^n k_i \int_a^b f_i$.
8. If $f \in \mathcal{R}[a, b]$ and $|f(x)| \leq M$ for all $x \in [a, b]$, show that $\left|\int_a^b f\right| \leq M(b - a)$.
9. If $f \in \mathcal{R}[a, b]$ and if $(\dot{\mathcal{P}}_n)$ is any sequence of tagged partitions of $[a, b]$ such that $\|\dot{\mathcal{P}}_n\| \to 0$, prove that $\int_a^b f = \lim_n S(f; \dot{\mathcal{P}}_n)$.
10. Let $g(x) := 0$ if $x \in [0, 1]$ is rational and $g(x) := 1/x$ if $x \in [0, 1]$ is irrational. Explain why $g \notin \mathcal{R}[0, 1]$. However, show that there exists a sequence $(\dot{\mathcal{P}}_n)$ of tagged partitions of $[a, b]$ such that $\|\dot{\mathcal{P}}_n\| \to 0$ and $\lim_n S(g; \dot{\mathcal{P}}_n)$ exists.
11. Suppose that f is bounded on $[a, b]$ and that there exists two sequences of tagged partitions of $[a, b]$ such that $\|\dot{\mathcal{P}}_n\| \to 0$ and $\|\dot{\mathcal{Q}}_n\| \to 0$, but such that $\lim_n S(f; \dot{\mathcal{P}}_n) \neq \lim_n S(f; \dot{\mathcal{Q}}_n)$. Show that f is not in $\mathcal{R}[a, b]$.
12. Consider the Dirichlet function, introduced in Example 5.1.6(g), defined by $f(x) := 1$ for $x \in [0, 1]$ rational and $f(x) := 0$ for $x \in [0, 1]$ irrational. Use the preceding exercise to show that f is *not* Riemann integrable on $[0, 1]$.
13. Suppose that $c \leq d$ are points in $[a, b]$. If $\varphi : [a, b] \to \mathbb{R}$ satisfies $\varphi(x) = \alpha > 0$ for $x \in [c, d]$ and $\varphi(x) = 0$ elsewhere in $[a, b]$, prove that $\varphi \in \mathcal{R}[a, b]$ and that $\int_a^b \varphi = \alpha(d - c)$. [*Hint*: Given $\varepsilon > 0$ let $\delta_\varepsilon := \varepsilon/4\alpha$ and show that if $\|\dot{\mathcal{P}}\| < \delta_\varepsilon$ then we have $\alpha(d - c - 2\delta_\varepsilon) \leq S(\varphi; \dot{\mathcal{P}}) \leq \alpha(d - c + 2\delta_\varepsilon)$.]
14. Let $0 \leq a < b$, let $Q(x) := x^2$ for $x \in [a, b]$ and let $\mathcal{P} := \{[x_{i-1}, x_i]\}_{i=1}^n$ be a partition of $[a, b]$. For each i, let q_i be the positive *square root* of
$$\tfrac{1}{3}\left(x_i^2 + x_i x_{i-1} + x_{i-1}^2\right).$$
   (a) Show that q_i satisfies $0 \leq x_{i-1} \leq q_i \leq x_i$.
   (b) Show that $Q(q_i)(x_i - x_{i-1}) = \frac{1}{3}\left(x_i^3 - x_{i-1}^3\right)$.
   (c) If $\dot{\mathcal{Q}}$ is the tagged partition with the same subintervals as $\mathcal{P}$ and the tags q_i, show that $S(Q; \dot{\mathcal{Q}}) = \frac{1}{3}\left(b^3 - a^3\right)$.
   (d) Use the argument in Example 7.1.4(c) to show that $Q \in \mathcal{R}[a, b]$ and
$$\int_a^b Q = \int_a^b x^2 dx = \tfrac{1}{3}\left(b^3 - a^3\right).$$
15. If $f \in \mathcal{R}[a, b]$ and $c \in \mathbb{R}$, we define g on $[a + c, b + c]$ by $g(y) := f(y - c)$. Prove that $g \in \mathcal{R}[a + c, b + c]$ and that $\int_{a+c}^{b+c} g = \int_a^b f$. The function g is called the c-**translate** of f.